Fiscal Year:	FY 2011	**Task Last Updated:**	FY 06/13/2011
PI Name:	Mitchell, Cary Ph.D.		
Project Title:	Maturing Technology for Major Reductions in Energy, Mass, and Crew Time for Plant Lighting in Space		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Plant Biology		
Space Biology Cross-Element Discipline:	(1) Reproductive Biology		
Space Biology Special Category:	(1) Bioregenerative Life Support		
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City:	West Lafayette	**State:**	IN
Zip Code:	47907-2010	**Congressional District:**	4
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2008 Space Biology--Microbial, Plant, Cell Biology: NNH08ZTT003N
Start Date:	06/09/2009	**End Date:**	07/31/2013
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	2
No. of Master's Candidates:	3	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA KSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: new end date is 7/31/2013 per NSSC, and also is now showing start date of 6/9/2009 (Ed., 6/11/12) NOTE: new end date is 7/31/2012 per NSSC (Ed. 5/3/2011) NOTE: change in Period of Perfomance to 8/1/2009-7/31/2011 (from 6/9/2009-7/8/2011), per NSSC information (Ed., 3/11/2011)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX09AL99G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The high Equivalent System Mass (ESM) of electric lighting forced NASA to put use of plants on hold for space life-support systems. Significant reductions in ESM are required for loop closure to occur. A novel solution to the problem is proposed that combines solid-sate lighting technology, non-traditional light distribution, "smart"sensing and switching to avoid wasting energy, and optimization of lighting and related environmental parameters for photosynthesis of fresh salad crops. Clusters of light-emitting diodes (LEDs) that are air-cooled to remove waste heat from the back of emitters will be positioned overhead close to the surfaces of lettuce and strawberry crops. The favorable surface temperature of emitters will save substantial energy by distributing adequate light levels in close proximity to leaves without scorching them. Emission spectra of LEDS will be selected to match absorption spectra of leaves. At intervals, red and blue LEDs will be switched off and green LEDs will be flashed in patterns to test the proximity of leaves to each cluster of LEDs on the light bank. Since green light is not absorbed as efficiently as red or blue, photodiodes mounted on the light bank in the vicinity of LED clusters will detect green reflectance, the intensity of which depends upon proximity of reflecting leaves to the LEDs. Only clusters within an effective distance from green-reflecting leaves will switch on red and blue LEDs, so that energy will not be wasted lighting empty spaces between plants. Minimum light intensities and appropriate R/B spectral ratios will be identified that give adequate photosynthetic rates by crop stands, especially in combination with CO2 enrichment and non-limiting temperatures. These technical advancements in lighting will enable the energetic affordability of a "salad machine" at the Lunar outpost, leading to future loop closure for space life-support systems. The task is to develop and demonstrate proof of concept for two novel technologies that together will greatly advance the potential for growing food crops in space for crew life support at distant space destinations. The first technology is a “smart” LED-based sole-source lighting system for saving considerable energy for many reasons. The second technology is a crop-stand gas-exchange system consisting of a large cuvette, an open gas-exchange system, and computer controls for CO2, temperature, and atmosphere flow rate. Lighting parameters are controlled by a second computer. Using the smart lighting system in conjunction with the crop gas-exchange system will give real-time crop response of hydroponically grown crops to light (PPF, R/B ratio), temperature, and CO2 at all stages of crop production, which will define additional opportunities for significant energy savings during controlled-environment crop production.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The project has broad impact for life on Earth as well as in space. On Earth, controlled environment agriculture (CEA) has limitations of profitability, economic viability, and affordability parallel to the issues that hold back the feasibility of using plants for crew life support in space. Interestingly, the single factor that limits the widespread use of CEA on Earth also is that of crop lighting! Supplemental and sole-source lighting using traditional high-intensity discharge (HID) lamps requires major electrical power consumption for waste-heat rejection as well as for lighting per se. The replacement of hot, inefficient HID lamps with solid-state light-emitting diode (LED) technology will benefit CEA and plant research for many reasons: LEDs are small, require much less mass and volume than HIDs, do not require massive ballasts, are extremely long-lived, are relatively cool at photon-emitting surfaces because waste heat is rejected elsewhere, emission wavelengths can be selected to match plant pigments, and can be operated at much lower power levels than HIDs because they can be placed in close proximity to plant tissues. As well, the field of LED technology is progressing at an ever-increasing rate, and electric-to-light conversion efficiencies are increasing at a rapid rate for LEDs. There is great interest in the use of LEDs for CEA on Earth as well as for crew life support in space. In addition, this project addresses the development of “smart” LED lighting systems that automatically sense plant position and size, and subsequently energize LEDs only directly above or in line with plant tissues. When used in conjunction with and supplemental to solar light in greenhouses on Earth, smart LED lighting systems also will monitor the instantaneous contribution of solar light as it tracks across greenhouses daily, and will automatically adjust LED switching and/or dimming/brightening controls in response to the total light environment. The crop gas-exchange system that is part of this project will be used in conjunction with smart LEDs to determine day-to-day crop tolerances, “preferences,” and limits for intensity of light, quality of light, and interaction of light with other important environmental parameters such as CO2 and temperature at all stages of crop development. This capability will greatly accelerate the optimization of environmental conditions needed to produce crops rapidly, efficiently, and affordably, which is just as important for CEA on Earth as it is for closed-loop life support in space.
	The first reporting period of the project period focused on the development of a gas-exchange, cuvette-based system and LED lighting hardware. Engineers from the Orbital Technologies Corporation (ORBITEC, Madison, WI) developed two overhead LED smart lighting systems and associated control software. Ideal Engineering Solutions (Lafayette, IN) designed and fabricated a custom gas-exchange cuvette enclosure for hydroponic crop growth to allow environmental and atmosphere control. All designs were made with input from the PI and communications between subcontractors. A graduate student in Mechanical Engineering Technology at Purdue (K. Spence) integrated structural aspects of the cuvette including the hydroponic system and the atmosphere control feedback system, while a Purdue graduate student in Electrical Engineering Technology (B. Riggs) automated the electronic aspects of the system and developed control and monitoring software. The second reporting period has focused on testing and troubleshooting of the high-efficiency-lighting-with-adaptive-control (HELIAC) smart lighting system and custom cuvette enclosure. A Senior undergraduate student in Electrical Engineering Technology at Purdue (I. Traylor) focused on the cuvette gas-exchange system, while, a Purdue graduate student in Aerospace Engineering (L. Poulet), Research Scientist (G. Massa), and undergraduate in Horticulture (M. Dzakovich) have been troubleshooting the overhead HELIAC lighting system while working with plants. Leaf lettuce is the model plant system for the proof-of-concept tests. HELIAC flashes light from green LEDs onto a crop stand in the dark. Light reflecting back from green leaves to photodiodes on the LED array detects plant position and size, signalling switching on of an appropriate numbers of red and blue LEDs in appropriate positions above the crop stand. For HELIAC version 2, the main issue has been that small lettuce plants (four-leaf stage and under) are not detected, whereas larger plants are detected with increasing ease as they grow. This limitation is due to false-positive detection “noise” induced by structural crossbars and frames of the LED panel. Small plants were not detected because the intensity of their green reflection, even at the maximum flashing intensity, was less than that reflected by the crossbars. Thus, the system detected the structural bars as main objects, and small plants as noise. For a given detection sensitivity (intensity + threshold), the software eliminates a certain level of noise, including signals from small plants. To avoid this problem, a mirrored-foil collar was placed around the base of each plant. Different mirror diameters were tested. Increasing mirror diameters were correlated with improving plant detection. HELIAC must be able to detect individual plants separately when they are small, without background noise, and follow them through their growth curve until vegetation populates the entire growing surface below the array, at

	which time, and only then, all LEDs on the array should be energized. If each plant has to be surrounded by a large collar of 7 cm diameter, which gave excellent small-plant detection, the whole LED panel would be energized all the time and discriminating, smart lighting would never be achieved. The HELIAC hardware and software were returned to ORBITEC for 3 months during the reporting period for refurbishment and maintenance. A number of improvements were made, including increasing flashing intensity 30% and removing collimating lenses from the red LEDS, which allowed closer placement of the array to crop surfaces with good mixing of red + blue hues. A dark-scan feature was added to the software. The growing surface is first scanned without plants, and the user then adds the plants on the growing surface and reruns the detection. The dark-scan signals are then deleted from the plant scan, keeping only signals reflected by the plants. The dark-scan feature is now being tested with small plants, with and without small reflective collars. Each detection run is now saved as both a picture and a text file, which helps with data management.
Task Progress:	The goal of improving mechanical aspects of the crop-stand gas-exchange system during the past reporting period has focused primarily on the large crop-stand cuvette. A heater/cooler and humidification column were installed in/on the cuvette, respectively, and the integrated gas-exchange system was tested to determine if differential CO2 levels could be accurately measured and are essentially zero between air passing through the cuvette vs. a cuvette bypass without plants growing in the cuvette. The gas-exchange system needs to permit accurate measurement of differences between sample and reference CO2 levels once sample air has passed through and exited the cuvette. Theoretically, the differential between sample and reference gasses after passing through an empty cuvette should be 0 PPM. However, when a known concentration of 1000 PPM CO2 was pumped through the system, error as large as -200 PPM has been measured in the sample stream relative to the reference stream. Suspected problems with the cuvette include the following: leaks, too large an internal volume preventing the cuvette from operating under slight positive pressure, erratic gas mixing near the outlet port, “unprocessed” air short circuiting between cuvette inlet and outlet ports or bypassing the “processing” space immediately above the crop stand, and temperature artifacts. Dry ice and hot water were used to create a white vapor inside the cuvette and also to generate a positive internal pressure so that leakage points could be visibly identified. Several leaks along the base of the cuvette were discovered and were subsequently sealed with silicone. Under normal cuvette operation, such leaks might draw room air into the cuvette if it is not maintained under sufficient positive pressure due to a large pressure drop as inlet air enters the large cuvette volume. The working head-space volume of the cuvette will be reduced to that occurring between the bottom of the height-adjustable LED array and the fixed hydroponic container. A flexible barrier will be created consisting of polyethylene film. This adaptation will ensure that all inlet air will pass across the space between the plants and light source. Reducing the working head-space volume of the cuvette should also help maintain a slight positive pressure within the cuvette, which in turn will help eliminate contamination from room air through undetected leaks. When the Peltier temperature-control unit is active during testing, error as high as -200PPM CO2 was measured. Testing the cuvette with the Peltier unit switched off yielded results that were ±10PPM of expected, indicating that the Peltier unit is a large source of the error. The Peltier unit operates by adjusting the temperature of metal fins, then blowing cuvette air across those fins to warm or cool air to the desired temperature. However, the way it presently is configured, inlet air can be short-circuited toward the outlet port and exit the cuvette along with sample air to be processed photosynthetically by plants. The cuvette design either will be modified so that it only directs downstream air in one direction or is moved to a different location in the cuvette. Another possibility is that the proximity of the Peltier unit to the outlet port is causing a temperature differential between outlet air and cuvette by-pass (reference) air, and this is creating an apparent CO2 concentration difference in the differential IRGA. This possibility reinforces the need to ensure that air cooled by the Peltier unit is directed only to the inlet side of the headspace above the plants that is being irradiated and heated by the LED array. The room-temperature inlet air must be brought to the cuvette temperature set point at the level of the plant canopy by the time it passes through or over the plant canopy. Any air cooled by the Peltier unit will have to be diverted away from the outlet port only to the plant-growth area of the cuvette. An alternative, simpler solution will be to create a new outlet port in the channel space just past the crop canopy below the flexible membrane point of attachment and seal off the existing outlet port. These options will be systematically implemented and evaluated to get the cuvette functioning early in the next reporting period so that plant testing of plant gas-exchange analysis in combination with HELIAC can be completed.

Bibliography Type:	Description: (Last Updated: 04/16/2020)
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